

Reinforcement learning.



Overview: Reinforcement Learning $\rightarrow$ Goal: finding the most optimal policy

Concepts:

Markov decision process

Bellman equations + optimality.

Algorithms & methods

Value iteration, policy iteration, truncated policy iteration.

Trade between a policy and value

Monte-Carlo learning: model free

Bellman equations:

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1} \xrightarrow{A_{t+1}} R_{t+2}, S_{t+2} \dots$$

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$G_t = R_{t+1} + \gamma (R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \dots)$$

$$= R_{t+1} + \gamma G_{t+1}$$

State value: An average reward of all possible trajectories w/ initial state S .

$$V_{\pi}(S) = E[G_t | S_t = S]$$

$$= E[R_{t+1} + \gamma G_{t+1} | S_t = S]$$

$$= E[R_{t+1} | S_t = S] + \gamma E[G_{t+1} | S_t = S]$$

$$E[R_{t+1} | S_t = S] = E[E[R_{t+1} | S_t = S, A_t = a] | S_t = S]$$

$$= \sum_a \pi(a|S) E[R_{t+1} | S_t = S, A_t = a]$$

$$= \sum_a \pi(a|S) \sum_{\mathbf{r}} P(\mathbf{r} | S, a) r$$

$$E[G_{t+1} | S_t = S] = \gamma \int E[G_{t+1} | S_t = S, S_{t+1} = S', G_{t+1} = G'] | S_t = S]$$

$$= \gamma \int E[G_{t+1} | S_{t+1} = S'] P(S' | S_t = S)$$

$$= \gamma \int_{S'} V_{\pi}(S') P(S' | S)$$

$$= \gamma \sum_{S'} V_{\pi}(S') \sum_a P(S' | S, a) \pi(a|S)$$

$$= \gamma \sum_{S'} V_{\pi}(S') P(S' | S) \sum_a \pi(a|S)$$

$$\Rightarrow V_{\pi}(S) = E[R_{t+1} | S_t = S] + \gamma E[G_{t+1} | S_t = S]$$

$$= \sum_a \pi(a|S) \sum_{\mathbf{r}} P(\mathbf{r} | S, a) r + \gamma \sum_{S'} V_{\pi}(S') P(S' | S) \sum_a \pi(a|S)$$

$$V_{\pi}(S) = \sum_a \pi(a|S) \left[\sum_{\mathbf{r}} P(\mathbf{r} | S, a) r + \gamma \sum_{S'} V_{\pi}(S') P(S' | S, a) \right] \Rightarrow \text{Bellman equation}$$

Matrix-vector form: $V_{\pi}(S_t) = r_{\pi}(S_t) + \sum P_{\pi}(S_j | S_t) V_{\pi}(S_j)$

$$V_{\pi} = r_{\pi} + \gamma P_{\pi}^T V_{\pi}$$

$$V_{\pi} = \begin{pmatrix} V_{\pi}(S_1) \\ \vdots \\ V_{\pi}(S_n) \end{pmatrix}$$

$$\begin{bmatrix} v_{\pi}(s_1) \\ v_{\pi}(s_2) \\ v_{\pi}(s_3) \\ v_{\pi}(s_4) \end{bmatrix} = \begin{bmatrix} r_{\pi}(s_1) \\ r_{\pi}(s_2) \\ r_{\pi}(s_3) \\ r_{\pi}(s_4) \end{bmatrix} + \gamma \underbrace{\begin{bmatrix} p_{\pi}(s_1|s_1) & p_{\pi}(s_2|s_1) & p_{\pi}(s_3|s_1) & p_{\pi}(s_4|s_1) \\ p_{\pi}(s_1|s_2) & p_{\pi}(s_2|s_2) & p_{\pi}(s_3|s_2) & p_{\pi}(s_4|s_2) \\ p_{\pi}(s_1|s_3) & p_{\pi}(s_2|s_3) & p_{\pi}(s_3|s_3) & p_{\pi}(s_4|s_3) \\ p_{\pi}(s_1|s_4) & p_{\pi}(s_2|s_4) & p_{\pi}(s_3|s_4) & p_{\pi}(s_4|s_4) \end{bmatrix}}_{P_{\pi}} \begin{bmatrix} v_{\pi}(s_1) \\ v_{\pi}(s_2) \\ v_{\pi}(s_3) \\ v_{\pi}(s_4) \end{bmatrix}$$

$$V_{\pi} = r_{\pi} + \gamma P_{\pi}^T V_{\pi}$$

Solving gives:

$$V_{\pi} = (I - \gamma P_{\pi}^T)^{-1} r_{\pi}$$

Action values vs State value

Action:

$$Q_{\pi}(S, a) = E[G_t | S_t = S, A_t = a]$$

$$= \int P(\mathbf{r} | S, a) r + \gamma \sum_{S'} V_{\pi}(S') P(S' | S, a)$$

State: $E[G_t | S_t = S] = \int_a E[G_t | S_t = S, A_t = a] \pi(a|S)$

$$= \int_a Q_{\pi}(S, a) \pi(a|S)$$

Regular Bellman eq.

$$V_{\pi}(S) = \sum_a \pi(a|S) \left[\int P(\mathbf{r} | S, a) r + \gamma \sum_{S'} V_{\pi}(S') P(S' | S, a) \right]$$

Bellman optimality equation:

$$V(S) = \max_a \sum_{\pi} \pi(a|S) \left[\int P(\mathbf{r} | S, a) r + \gamma \sum_{S'} V_{\pi}(S') P(S' | S, a) \right]$$

$$= \max_a \sum_{\pi} \pi(a|S) Q_{\pi}(S, a)$$

$$= \max_{a \in A} \{ Q_{\pi}(S, a) \}$$

$$\hat{a} = \operatorname{argmax}_a Q_{\pi}(S, a)$$

$$\pi(a|S) = \begin{cases} 1 & \text{if } a = \hat{a} \\ 0 & \text{otherwise} \end{cases}$$

Matrix-vector form:

$$V = \max_a (r_a + \gamma P_a^T V)$$

Value Iteration, Policy Iteration.
Truncated policy eg.

* Value Iteration
Bellman optimality equation
 $V = f(V) = \max_a (r_a + \gamma \sum_{s'} P_{sa}(s') V(s'))$

Step 1: Policy update.

$$\pi_{k+1} = \arg\max_{\pi} \{ \pi(s) \left(\sum_{s'} P(s'|s,a) r + \gamma \sum_{s'} P(s'|s,a) V_k(s') \right) \} \quad \forall s \in S$$

$$\pi_{k+1}(a|s) = \begin{cases} 1 & a = a_k^*(s) \\ 0 & \text{otherwise} \end{cases}$$

$$a_k^* = \arg\max_a q_k(s,a)$$

Step 2: Value update

$$V_{k+1} = r_{\pi_{k+1}} + \gamma \sum_{s'} P_{\pi_{k+1}}(s|s,a) V_k(s') \Rightarrow V_{k+1}(s) = \sum_a \pi_{k+1}(a|s) \left[\sum_{s'} P(s'|s,a) r + \gamma \sum_{s'} P(s'|s,a) V_k(s') \right] \quad \forall s \in S$$

$$V_{k+1} = \max_a q_k(s,a)$$

Pseudo code
 initialize V_0 .
 if $\|V_k - V_{k-1}\| < \epsilon$, $k \leq 0$
 for $s = 1, 2, \dots, k \in S$
 for $a \in A$

compute $q_k(s,a) = \sum_{s'} P(s'|s,a) r + \gamma \sum_{s'} P(s'|s,a) V_k(s')$
 compute $a_k^* = \arg\max_a q_k(s,a)$
 $\pi_{k+1}(a|s) = \begin{cases} 1 & a = a_k^* \\ 0 & \text{otherwise} \end{cases}$ (Policy update)
 $V_{k+1} = \max_a q_k(s,a)$ (Value update)

Policy Iteration:
 start w/ policy π_0

Step 1: Policy evaluation:
 $V_{\pi_k} = r_{\pi_k} + \gamma P_{\pi_k} V_{\pi_k}$ until convergence. $V_{\pi_k}^{(j+1)}(s) = \sum_a \pi_k(a|s) \left(\sum_{s'} P(s'|s,a) r + \gamma \sum_{s'} P(s'|s,a) V_{\pi_k}^{(j)}(s') \right) \quad \forall s \in S$
 stop when $j \rightarrow \infty$

Step 2: Policy improvement

$\pi_{k+1} = \arg\max_{\pi} (r_{\pi} + \gamma P_{\pi} V_{\pi_k}) \Rightarrow \pi_{k+1}(s) = \arg\max_{\pi} \sum_a \pi(a|s) \left(\sum_{s'} P(s'|s,a) r + \gamma \sum_{s'} P(s'|s,a) V_{\pi_k}(s') \right) \quad \forall s \in S$

$$= \arg\max_a \sum_a \pi(a|s) q(s,a)$$

$$a_k^*(s) = \arg\max_a q(s,a)$$

$$\pi_{k+1}(a|s) = \begin{cases} 1 & \text{if } a = a_k^*(s) \\ 0 & \text{otherwise} \end{cases}$$

Algorithm:

Initialize w/ policy π_0 .
 Start w/ an initial guess V_{π_0}
 $V_{\pi_0}^{(0)} = \sum_a \pi_0(a|s) \left[\sum_{s'} P(s'|s,a) r + \gamma \sum_{s'} P(s'|s,a) V_{\pi_0}^{(0)}(s') \right]$ } Policy evaluation
 $\forall s \in S$
 $\forall a \in A$
 $q(s,a) = \sum_{s'} P(s'|s,a) r + \gamma \sum_{s'} P(s'|s,a) V_{\pi_k}(s')$
 $a_k^*(s) = \arg\max_a q(s,a)$
 $\pi_{k+1}(a|s) = \begin{cases} 1 & \text{if } a = a_k^*(s) \\ 0 & \text{otherwise} \end{cases}$ } Policy improvement

Marko (also learning: model free)

$E[x] = \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$

convert policy iteration to model free.

Remarks:

Policy evaluation: $V_{\pi_k} = r_{\pi_k} + \gamma P_{\pi_k} V_{\pi_k}$
 improvement: $\pi_{k+1} = \arg\max_{\pi} (r_{\pi} + \gamma P_{\pi} V_{\pi_k})$

$$\pi_{k+1} = \arg\max_{\pi} \sum_a \pi(a|s) q(s,a) \quad \forall s \in S$$

$q_{\pi_k}(s,a) = E[r_t + \gamma V_{\pi_k}(s_t + a, A_t = a)] \approx \frac{1}{n} \sum_{i=1}^n g^i(s,a)$

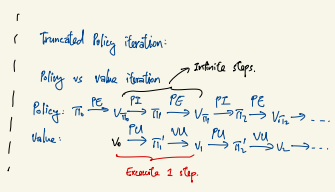
MC-basic algorithm

$\forall s \in S$
 $\forall a \in A$

select n $g^{(i)}(s,a)$
 compute $q_{\pi_k}(s,a) = \frac{1}{n} \sum_{i=1}^n g^{(i)}(s,a)$ } Monte Carlo method / model free.

$a_k^*(s) = \arg\max_a q_{\pi_k}(s,a)$

$\pi_{k+1}(a|s) = \begin{cases} 1 & a = a_k^*(s) \\ 0 & \text{otherwise} \end{cases}$ } Policy improvement



Algorithm: (similar to policy iteration)

Initialize w/ policy π_0 .
 Start w/ an initial guess V_{π_0}
 $V_{\pi_0}^{(0)} = \sum_a \pi_0(a|s) \left[\sum_{s'} P(s'|s,a) r + \gamma \sum_{s'} P(s'|s,a) V_{\pi_0}^{(0)}(s') \right]$ } Policy evaluation
 $\forall s \in S$ } stop when j is truncated. (not to infinity!)
 $\forall a \in A$
 $q(s,a) = \sum_{s'} P(s'|s,a) r + \gamma \sum_{s'} P(s'|s,a) V_{\pi_k}(s')$
 $a_k^*(s) = \arg\max_a q(s,a)$
 $\pi_{k+1}(a|s) = \begin{cases} 1 & \text{if } a = a_k^*(s) \\ 0 & \text{otherwise} \end{cases}$ } Policy improvement